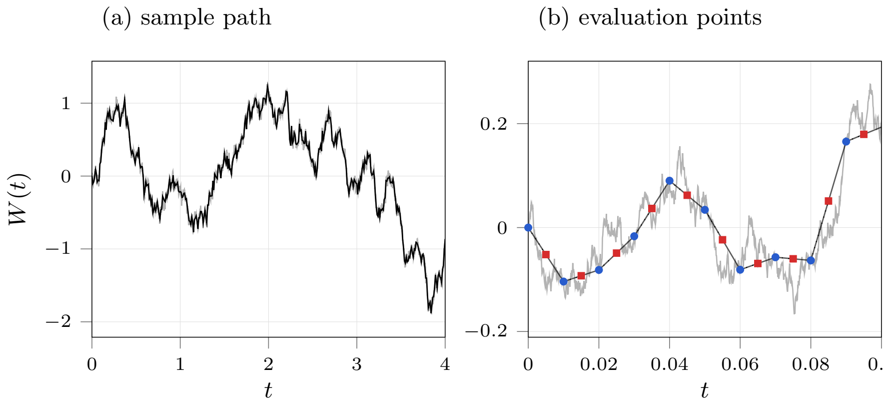

Figure 1: (a) A Brownian path at $\Delta t = 10^{-4}$ (pale) with the integration grid it is sampled on, $\Delta t = 10^{-2}$ (black), laid over it. (b) The first ten steps of that grid. The Itô rule evaluates the integrand at the left end of each step, $W(t_k)$ (blue circles); the Stratonovich rule takes the mean of the two ends, $\frac{1}{2}[W(t_k) + W(t_{k+1})]$ (red squares), which is the midpoint of the chord and not a value of the path. The dotted segments join each pair.

Itô versus Stratonovich

The two integrals differ only in where the integrand is evaluated inside a step. Figure 1 shows the sample path and the two evaluation points; figure 2 shows what the choice accumulates to.

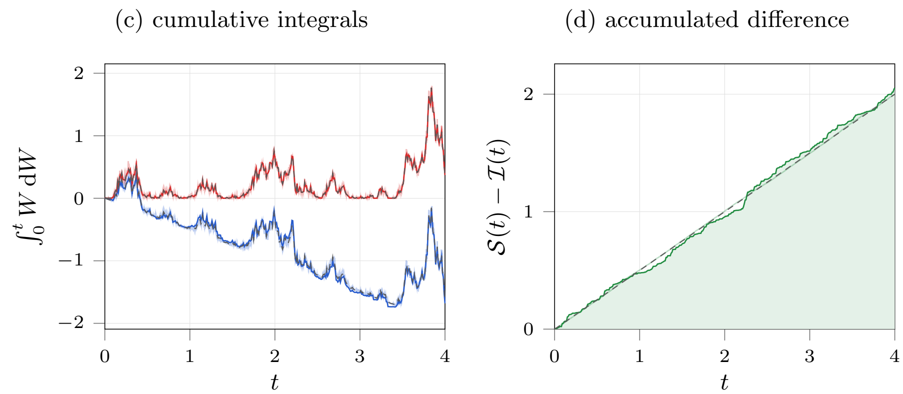

Figure 2: (c) $\int_0^t W\,\mathrm{d}W$ read as Itô (blue) and as Stratonovich (red), each drawn on the integration grid (thin) and, pale behind it, on the fine grid, against the closed forms $(W^2 - t)/2$ and $W^2/2$ (dashed). The Stratonovich sum reproduces the chain rule on any grid; the Itô sum sits off its closed form by $\frac{1}{2}(\sum \Delta W^2 - t)$. (d) The gap the two accumulate (green), against $t/2$ (dashed).